

Closure-Strain Geometry: Local Normal Forms and Conditional Matter Encodings

Peter Nero

Corrected seventh edition
July 2026

Abstract

We extract the exact geometry of closure strain from earlier Standard-Model interpretations. For an invertible rank-three comparison field, polar or Iwasawa reduction removes three orientation directions and leaves six strain directions. With a selected orthonormal flag these split orthogonally into scalar, traceless-diagonal, and shear sectors of dimensions $1 + 2 + 3$. We give explicit projectors and a norm identity, explain the relation and distinction between symmetric shear and the Heisenberg nil algebra, and state precisely what a closure Hessian proves. In particular, Hessian positivity does not select a unique Higgs, three families, Standard Model charges, confinement, mixing, or CP violation. Those identifications require representation, connection, action, and source theorems. Current q79 and finite-algebra calculations are incorporated at their declared profile-equivalence tier, and the q79-side finite bridge is sharpened: one universal flat differential line commutes with the $1 + 2 + 3$ lane projectors and lifts the normalized Reynolds Hessian exactly. This closes connection/holonomy and Hessian naturality at finite-symbol tier. The same-source continuum intertwiner from the local strain normal form to the nonzero-Chern physical HYM carrier remains isolated and open.

Revision note for this edition

Supersedes. *Closure-Strain Geometry: Local Normal Forms and Conditional Matter Encodings*, version 6.

Reason. Version 6 correctly left the physical local-to-q79 intertwiner open, but it predates the exact q79 universal-line and finite-Hessian square and therefore understates what is already closed on the target side.

Resolution. Version 7 retains the local $1 + 2 + 3$ theorem and adds the flat differential-line pullback, finite Reynolds projector, exact Hessian spectrum, and the precise curved-HYM nonpromotion guard.

Retained result. The local strain normal form and the executed embedded renormalized-SM profile branch both survive, as distinct results.

Remaining boundary. Their physical identification still requires the same-source metric, connection, and continuum-HYM intertwiner; strict no-knob branch and value selection remain stronger upgrades.

Contents

[Revision note for this edition](#)

1

1	How to Read Closure Strain	2
1.1	Stiffness is not yet mass	2
1.2	Argument map	3
2	Purpose and status	3
3	Comparison field and strain	3
4	Exact 1 + 2 + 3 normal form	4
5	Iwasawa interpretation and the nil warning	4
6	Closure cost and Hessian consequences	5
7	Conditional Higgs interpretation	5
8	Conditional matter and gauge encodings	6
9	Families, quarks, and order of breakdown	6
10	Yukawa magnitudes, mixing, and CP	6
11	The q79 connection problem	7
12	Status ledger	8
13	Conclusion	8

1 How to Read Closure Strain

Closure strain is best understood first as a local stiffness problem. The comparison field Q measures how two rank-three frames differ. Polar decomposition removes the orientation part, and $S = \log U$ records the remaining deformation. A closure functional $\mathcal{J}$ then assigns a cost to perturbing that aligned state.

The paper has three distinct achievements:

1. the local strain space has an exact flag-selected 1 + 2 + 3 orthogonal normal form;
2. the selected q79 target has an exact finite rank-1 + 2 + 3 carrier, shared line, and finite Hessian square; and
3. the finite Standard Model construction realizes matter and scalar profiles at its declared profile tier.

These results are mutually compatible, but they become one derivation only after a same-source bundle, connection, metric, and operator intertwiner is constructed.

1.1 Stiffness is not yet mass

For a finite model, write the quadratic action schematically as

$$\frac{1}{2}\dot{s}^T K \dot{s} - \frac{1}{2}s^T H s.$$

The closure Hessian H measures restoring cost. The physical squared frequencies or masses solve the generalized eigenvalue problem

$$Hv = m^2 Kv.$$

Rescaling a field changes the matrix entries of both H and K while leaving the physical generalized eigenvalues unchanged. Consequently a positive entry of H , or even a positive eigenvalue computed with an arbitrary coordinate norm, is not yet a mass prediction.

The same distinction applies to particle names. A one-dimensional scalar strain block is a geometric representation. To call it the Higgs requires a complex weak doublet representation, a selected kinetic term and potential, symmetry breaking, and the renormalized pole map. The local dimension count only makes such an intertwiner possible.

1.2 Argument map

Sections 2–6 establish the status, comparison, projector, Iwasawa, and Hessian geometry. Sections 7–10 state the independent Higgs, matter, family, Yukawa, mixing, and CP gates. Section 11 identifies what is already closed on the q79 target side and isolates the continuum source map still needed to join the local geometry to that target.

2 Purpose and status

The paper proves a local normal form and organizes possible matter encodings. It does not derive the Standard Model from the normal form. We distinguish:

Local theorem. Linear algebra and differential geometry following from the stated rank-three comparison field and selected flag.

Selected realization. The q79 trace carrier and the finite $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ branch supplied by independent packets.

Profile equivalence. Reproduction of the embedded renormalized Standard Model using the declared one-shared-physical-primitive/profile standard.

Strict selection. Derivation of the observed branch and numerical data without target-profile selection; this remains a stronger program.

3 Comparison field and strain

Let $TP, TI \rightarrow B$ be oriented Euclidean rank-three bundles and let

$$Q = Q_{\text{WW}} \in \Gamma(\text{Hom}(TP, TI))$$

be invertible and orientation preserving on an admissible domain. In local orthonormal frames,

$$Q = RU, \quad R \in SO(3), \quad U = (Q^T Q)^{1/2} > 0.$$

Definition 3.1 (Logarithmic closure strain). The logarithmic strain is

$$S = \log U = \frac{1}{2} \log(Q^T Q) \in \text{Sym}(3, \mathbb{R}).$$

It is dimensionless. Any physical dimension assigned to a fluctuation of S must come from a separately normalized action and field coordinate.

The nine local comparison components split as

$$\text{Mat}(3, \mathbb{R}) = \mathfrak{so}(3) \oplus \text{Sym}(3, \mathbb{R}), \quad 9 = 3 + 6.$$

This is a local field-component decomposition, not a product decomposition of a manifold.

4 Exact 1 + 2 + 3 normal form

Choose an orthonormal flag. For $S \in \text{Sym}(3, \mathbb{R})$ write

$$\text{Diag}(S) = \text{the diagonal matrix with the diagonal of } S$$

and define

$$P_{\text{sc}}S = \frac{\text{tr } S}{3}I_3, \quad P_{\text{sh}}S = \text{Diag}(S) - P_{\text{sc}}S, \quad P_{\text{nil}}S = S - \text{Diag}(S).$$

Theorem 4.1 (Closure-strain projector theorem). *The three maps are self-adjoint orthogonal projectors for the Frobenius inner product. Their images have dimensions 1, 2, 3, and*

$$S = P_{\text{sc}}S + P_{\text{sh}}S + P_{\text{nil}}S, \quad \|S\|_F^2 = \|P_{\text{sc}}S\|_F^2 + \|P_{\text{sh}}S\|_F^2 + \|P_{\text{nil}}S\|_F^2.$$

Proof. The images are, respectively, scalar diagonal matrices, traceless diagonal matrices, and symmetric zero-diagonal matrices. Their pairwise Frobenius inner products vanish. Each displayed map is the identity on its image and zero on the other two images, proving idempotence and self-adjointness. Their dimensions are one, two, and three, which sum to $\dim \text{Sym}(3) = 6$. $\square$

Remark 4.2 (Dependence on the flag). The trace line is canonical, but the split of the traceless five-dimensional sector into dimensions two and three depends on the selected flag. Under the full $SO(3)$ action the invariant decomposition is 1 + 5. Therefore an MTT claim using 1 + 2 + 3 must identify the operator or geometry selecting the flag.

5 Iwasawa interpretation and the nil warning

For $Q \in GL^+(3, \mathbb{R})$, QR/Iwasawa decomposition gives

$$Q = KAN, \quad K \in SO(3),$$

where A is positive diagonal and N is upper unitriangular. Their dimensions are

$$\dim K = 3, \quad \dim A = 3 = 1 + 2, \quad \dim N = 3.$$

The Lie algebra of N consists of strictly upper-triangular matrices and is the three-dimensional Heisenberg nil algebra.

The symmetric off-diagonal space $\text{im } P_{\text{nil}}$ is also three-dimensional, but it is not itself that nil Lie algebra: symmetric off-diagonal matrices are not closed under commutators. The two spaces are linearly identified at a chosen background through the differential of the polar/Iwasawa coordinate change. A paper using the word “nil” must state whether it means the Heisenberg group, its Lie algebra, a symmetric shear coordinate, or a boundary/anchoring operator.

This resolves the valid core of the circle–lens–nil picture: it can describe a phase line, a finite/projective transport lane, and a triangular anchoring lane. It does not prove a literal nested or product topology.

6 Closure cost and Hessian consequences

Let $\mathcal{J}$ be a C^3 gauge-invariant closure functional on a Sobolev neighborhood of an aligned configuration S_* , and suppose $D\mathcal{J}(S_*) = 0$. Its quadratic variation on a gauge-fixed slice is

$$\mathcal{J}(S_* + s) = \mathcal{J}(S_*) + \frac{1}{2}\langle s, Hs \rangle + R_3(s), \quad |R_3(s)| \leq C\|s\|^3.$$

Theorem 6.1 (What a positive strain Hessian proves). *If H is self-adjoint and $H \geq cI$ on the six-dimensional strain slice for some $c > 0$, then S_* is a strict local minimum on that slice and all six quadratic strain directions have positive restoring cost. Positivity alone does not select one of the six directions as a physical scalar and does not identify any eigenvalue with a pole mass.*

Proof. Taylor's theorem and the coercive quadratic bound imply strict local minimality for sufficiently small s . A positive operator can have between one and six distinct eigenspaces and carries no representation or kinetic normalization data. Pole masses depend on the canonically normalized kinetic operator as well as H . $\square$

If H commutes with all three projectors, it has block form

$$H = H_{\text{sc}} \oplus H_{\text{sh}} \oplus H_{\text{nil}}.$$

This block invariance is an additional theorem or symmetry assumption. A generic Hessian mixes the three selected sectors.

Even in the block-diagonal case, the spectrum answers a stability question before it answers a particle question. The eigenvectors must transform in the required gauge representations, survive the constraints, and be canonically normalized by the kinetic operator. Only then can an eigenvalue be transported to a pole observable.

7 Conditional Higgs interpretation

The trace line in P_{sc} supplies one local scalar coordinate. To identify it with a Higgs degree of freedom one must additionally prove:

1. the selected finite representation is an $SU(2)$ complex doublet with the required hypercharge, rather than a real singlet strain coordinate;
2. the alignment projector selects one rank-four real doublet module and removes or lifts all other scalar modules;
3. the kinetic term is positive and canonically normalized;
4. the potential has the required vacuum and symmetry-breaking orbit; and
5. the renormalized pole observable follows after threshold and RG transport.

In the selected finite-algebra computation, the unrestricted real fluctuation space contains three rank-four scalar-doublet modules. The q79/proto-spinor alignment rule is represented by a rank-four projector that removes eight unwanted real scalar directions at the declared profile tier. This is a substantive downstream selection result. It is not implied by Hessian positivity, and its identification with the local trace line still needs the same-source intertwiner.

8 Conditional matter and gauge encodings

The selected finite branch uses

$$\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$$

on an explicit three-family particle–antiparticle carrier. At the profile tier its finite Dirac operator satisfies the declared self-adjointness, grading, real-structure, order-zero, and order-one checks. The anomaly-free circle and the usual gauge quotient are recovered on that selected spectrum.

These results are downstream finite-algebra theorems. They do not follow from the dimensions 1, 2, 3 alone. In particular:

- rank one does not by itself select hypercharge normalization;
- rank two does not by itself select weak chirality or a quaternionic real structure;
- rank three does not by itself select color representations or confinement; and
- three carrier lanes do not by themselves prove three fermion families.

The exact representation, bimodule, grading, real structure, and anomaly calculation must remain visible wherever Standard Model language is used.

9 Families, quarks, and order of breakdown

The hypothesis that quarks represent a second-order closure breakdown can be encoded by assigning leptonic states to a first nontrivial quotient and quark states to an iterated or noncommuting response sector. To turn this into a theorem, one needs a filtration

$$0 \subset F_1 \subset F_2 \subset \mathcal{H}_{q79}$$

and a selected operator whose first response preserves F_1 while its second response reaches F_2/F_1 . The construction must reproduce representation content and source values without using the observed quark/lepton distinction as the selector. The present normal form makes such a theorem well-typed but does not prove it.

Family multiplicity is similarly an index or monodromy problem, not a consequence of the number of strain blocks. The selected $\mathbb{Z}_3$ family/character factor and the gauge-rank flag must remain separate tensor factors.

10 Yukawa magnitudes, mixing, and CP

A geometric Yukawa entry requires normalized zero modes and a trilinear functional, schematically

$$Y_{ijk} = \int_{X_6} \langle \psi_i, \mathcal{K}_H(\psi_j, h_k) \rangle \text{vol}_{X_6}.$$

Its value depends on the selected connection, metrics, normalization, Higgs mode, and transport scheme. Complex phase can arise from common-circle holonomy, but the holonomy class and orientation branch must be selected before comparison with CKM or PMNS data.

The current repositories close charged Yukawa magnitudes, the finite matrix, CKM profile, electroweak rows, and precision transport at the adopted embedded renormalized-SM/one-shared-physical-primitive/profile standard. This means the constructed finite operator is capable of the required SM profile and passes its frozen certificates. It does not mean the

six-dimensional strain normal form alone derives all measured values, nor does it close strict no-knob unique selection.

11 The q79 connection problem

For the selected degree-three cover,

$$\mathcal{H}_{q79} = L_{\text{shared}} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}), \quad \text{rank} = 1 + 2 + 3.$$

On the q79 branch complement, the unique nontrivial map

$$h_{S_3} : S_3 \rightarrow \mathbb{Z}_{64}, \quad h_{S_3}(\sigma) = 32 \epsilon(\sigma),$$

pulls the universal flat weight-one line over $B_{\nabla} \mathbb{Z}_{64}$ back to the SpinC determinant sign line for either admissible odd root. Denote this specified pullback by $(L_{\text{sh}}, \nabla_{\text{sh}})$.

Theorem 11.1 (Closed q79 finite target square). *The same line L_{sh} tensors all three q79 lanes and its scalar holonomy commutes with their projectors. For the selected two-copy sheet/edge symbol,*

$$P_{\text{Haar}} = \frac{1}{6} \sum_{g \in S_3} \rho(g), \quad H_{\text{fin}} = \kappa_{\text{fin}}(I - P_{\text{Haar}})$$

satisfy

$$\text{rank } P_{\text{Haar}} = 2, \quad \text{spec}(H_{\text{fin}}/\kappa_{\text{fin}}) = \{0^{\times 2}, 1^{\times 4}\}, \quad H_{\text{TT}} = \kappa_{\text{fin}} I_2.$$

The lifted operator is exactly $I_{L_{\text{sh}}} \otimes H_{\text{fin}}$. Thus the q79 connection/holonomy line and finite Hessian square are closed without a dimensionless fit.

Proof. The character identity $\chi_r \circ h_{S_3} = \text{sgn}$ gives the parallel determinant-line pullback. Scalar holonomy commutes with every sheet operator. Haar averaging is an orthogonal projector, and decomposition of the selected representation gives the displayed ranks and spectrum. $\square$

This theorem closes the target-side finite square, not the physical bridge. The flat root-stack line cannot equal the nonzero-Chern physical HYM connection. Likewise, it supplies no map from the local world-in-world strain bundle.

What has been gained is a fixed codomain for the missing map. The target projectors, line action, and normalized finite Hessian no longer have to be guessed while constructing the intertwiner. The remaining task is to show that the physical continuum geometry emits those already fixed objects, not merely another rank-six model with a similar spectrum.

The local strain bundle has the same rank profile. The needed map is

$$\mathfrak{J} : \text{im } P_{\text{sc}} \oplus \text{im } P_{\text{sh}} \oplus \text{im } P_{\text{nil}} \longrightarrow \mathcal{H}_{q79}.$$

Theorem 11.2 (Requirements for physical promotion). *Rank matching promotes to a same-source physical identification only if $\mathfrak{J}$ is a global bundle isomorphism and, on the selected domains,*

$$\mathfrak{J}^* G_{\text{HYM}} = G_{\text{strain}}, \quad \mathfrak{J} \nabla^{\text{strain}} = \nabla^{\text{HYM}} \mathfrak{J}, \quad \mathfrak{J} H_{\text{strain}} = H_{q79} \mathfrak{J},$$

with analogous identities for the retarded and overlap kernels used to emit the finite source rows.

Proof. A physical identification must be independent of local trivialization and must preserve the structures used to define the action and observables. Bundle, metric, connection, and operator intertwining are therefore necessary. They are also sufficient to transport the declared quadratic and overlap calculations between the two descriptions. $\square$

Constructing this map is the decisive remaining foundation theorem for using the local closure-strain geometry as the source of the q79 numerical carrier. It is no longer necessary to reconstruct the q79 finite target square while doing so; that square is the closed codomain of the required intertwiner.

12 Status ledger

Object	Status	Meaning
$9 = 3 + 6$ comparison split	proved	local orientation/strain decomposition
$6 = 1 + 2 + 3$ strain split	proved with flag	exact orthogonal projectors
Iwasawa $SO(3)AN$ dimensions	proved	local/group normal form
q79 trace-split rank carrier	selected theorem	global rank $1 + 2 + 3$ carrier
q79 shared line and finite Hessian	proved at finite-symbol tier	connection, holonomy, projector, and Hessian square
Finite SM algebra and profile operator	closed at declared profile tier	embedded renormalized-SM equivalence
Unique Higgs from Hessian	not a theorem	requires representation and alignment source
Three families from three lanes	not a theorem	requires index/monodromy source
Local strain-q79 identification	open	requires same-source intertwiner
Strict no-knob value selection	stronger upgrade	not claimed here

13 Conclusion

Closure-strain geometry supplies a useful and exact six-dimensional local normal form. Its real achievement is the explicit $1 + 2 + 3$ decomposition and its compatibility target with the selected q79 carrier. The target is now stronger than a rank match: its common flat differential line and normalized finite Hessian square are exact. Standard Model organization becomes credible only when the finite representation and source packets are cited at their actual tier. The paper therefore replaces broad inevitability claims with one concrete bridge theorem capable of promoting the local geometry into the already executed numerical branch.

Computational Evidence and Reproducibility

The local closure-strain decomposition is proved in the paper. The selected q=79, finite-matrix, anomaly, HYM, and internal TT rows provide concrete realizations or compatibility checks; they do not turn the conditional matter interpretation into a source theorem. The remaining Standard Model packets are corpus-state context.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A13/gr_tt_support (*derived exact*).
Exact-branch internal TT support certificate; physical normalization remains open.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A11/q79_exact_audit (*derived exact*).
Executable q=79 exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT q=79 theorem on the selected exact branch.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] S. Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Academic Press, 1978.
- [2] A. Connes, *Noncommutative Geometry*, Academic Press, 1994.
- [3] P. Nero, *MTT Current True SM Closure Consolidated Ledger*, internal theorem and verification packet, 2026.
- [4] P. Nero, *q79 Universal Shared Differential Line and Finite-Operator Intertwiner*, executable theorem packet, 2026.